

Urinalysis, Complete With Microscopic Examination

Test Details

METHODOLOGY

Reagent strip and light microscopy

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Test Includes

Color; appearance; specific gravity; pH; protein; glucose; occult blood; ketones; leukocyte esterase; nitrite; bilirubin; urobilinogen; microscopic examination of urine sediment

Use

Detect abnormalities of urine; diagnose and manage renal diseases, urinary tract infection, urinary tract neoplasms, systemic diseases, and inflammatory or neoplastic diseases adjacent to the urinary tract

Limitations

Insufficient volume may limit the extent of procedures performed. Metabolites of Pyridium® may interfere with the dipstick reactions by producing color interference. High vitamin C intake may cause an underestimate of glucosuria, or a false-negative nitrite test. Survival of WBCs is decreased by low osmolality, alkalinity, and lack of refrigeration.

Custom Additional Information

MICROSCOPY:

Crystalluria is frequently observed in urine specimens stored at room temperature or refrigerated. Such crystals are diagnostically useful when observed in warm, fresh urine by a physician evaluating microhematuria, nephrolithiasis, or toxin ingestion.

In abundance, **calcium oxalate** and/or **hippurate crystals** may suggest ethylene glycol ingestion (especially if known to be accompanied by neurological abnormalities, appearance of drunkenness, hypertension, and a high anion gap acidosis). Large numbers of calcium oxalate crystals occur, as well, with acute renal failure following methoxyflurane anesthesia. Urine is usually supersaturated in calcium oxalate, often in calcium phosphate, and acid urine is often saturated in uric acid. Yet crystalluria is uncommon (in warm, fresh urine) because of the normal presence of crystal inhibitors, the lack of available nidus, and the time factor. When properly observed in fresh urine, crystals may provide a clue to the composition of renal stones even not yet passed, the nidus for such stones, or, as such, have been associated with microhematuria.

Leukocyturia may indicate inflammatory disease in the genitourinary tract, including bacterial infection, glomerulonephritis, chemical injury, autoimmune diseases, or inflammatory disease adjacent to the urinary tract such as appendicitis¹ or diverticulitis.

White cell casts indicate the renal origin of leukocytes, and are most frequently found in acute pyelonephritis. White cell casts are also found in glomerulonephritis such as lupus nephritis, and in acute and chronic interstitial nephritis. When nuclei degenerate, such leukocyte casts resemble renal tubular casts.

Red cell casts indicate renal origin of hematuria and suggest glomerulonephritis, including lupus nephritis. Red cell casts may also be found in subacute bacterial endocarditis, renal infarct, vasculitis, Goodpasture syndrome, sickle cell disease, and in malignant hypertension. Degenerated red cell casts may be called “**hemoglobin casts.**” Orange to red casts may be found with myoglobinuria as well.

Hyaline casts occur in physiologic states (eg, after exercise) and many types of renal diseases.

Renal tubular (epithelial) casts are most suggestive of tubular injury, as in acute tubular necrosis. They are also found in other disorders, including eclampsia, heavy metal poisoning, ethylene glycol intoxication, and acute allograft rejection.

Granular casts: Very finely granulated casts may be found after exercise and in a variety of glomerular and tubulointerstitial diseases. Coarse granular casts are abnormal and are present in a wide variety of renal diseases. “**Dirty brown**” **granular casts** are typical of acute tubular necrosis.

Waxy casts are found especially in chronic renal diseases, and are associated with chronic renal failure; they occur in diabetic nephropathy, malignant hypertension, and glomerulonephritis, among other conditions. They are named for their waxy or glossy appearance. They often appear brittle and cracked.

Fatty casts are found in the nephrotic syndromes generally, diabetic nephropathy, other forms of chronic renal diseases, and glomerulonephritis. The fat droplets originate in renal tubular cells when they exceed their capacity to reabsorb protein of glomerular origin. Their inclusions have the features and significance of oval fat bodies.

Broad casts originate from dilated, chronically damaged tubules or the collecting ducts. They can be granular or waxy. **Broad waxy casts** are called “renal failure casts.”

Sperm is **not** reported in routine urinalysis exams. **Spermatozoa** may be seen in male urine related to recent or retrograde ejaculation. For sperm reporting in males only, order Postejaculatory Urine Microscopic Examination for Sperm [133116] (/tests/133116/postejaculatory-urine-microscopic-examination-for-sperm).

Specimen Requirements

Specimen

Urine (random)

Volume

3-9 mL (Vacutainer® red/yellow urine transport tube) (preferred) **or** 2 mL or greater (non-preserved urine)

Minimum Volume

3 mL (Vacutainer® red/yellow urine transport tube) (preferred) or 2 mL (non-preserved urine) (**Note:** This volume does **not** allow for repeat testing.)

Container

Vacutainer® red/yellow urine transport tube (preferred) or urine bottle/urine transport tube (non-preserved)

Collection Instructions

A voided specimen is usually suitable. If the specimen is likely to be contaminated, a clean catch midstream specimen is desirable.

Storage Instructions

Vacutainer® red/yellow urine transport tube: room temperature for up to 72 hours

Urine bottle/urine transport tube (non-preserved): room temperature up to 2 hours and refrigerated up to 48 hours

Causes for Rejection

Quantity not sufficient for analysis; improper labeling; tubes with boric acid or tartaric acid (e.g., BD gray-top C&S tube); Labcorp red-top C&S tube; Boricon, Bortix, or Boricult yellow-top C&S tubes; urine transport tube with Triclosan; specimen received in Vacutainer® red/yellow urine transport tube more than 72 hours old; specimen received in urine cup stored at room temperature greater than 2 hours old; specimen received in urine cup refrigerated more than 48 hours old; frozen specimens; Vacutainer® red/yellow urine transport tube less than 3 mL or greater than 9 mL

Footnotes

1. Scott JH 3rd, Amin M, Harty JI. Abnormal urinalysis in appendicitis. *J Urol.* 1983 May; 129(5):1015. PubMed 6854742

(<http://www.ncbi.nlm.nih.gov/pubmed/6854742>)